

Geometry-consistent conservative transfer of potential-flow states to an embedded-boundary volume-of-fluid solver

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Abstract

Transferring a potential-flow state to a volume-of-fluid (VOF) solver requires accurate liquid-cell velocities on the geometry used by the receiver. We reconstruct liquid-cell moments by integrating a harmonic streamfunction over the polygonal cell boundary. For thin strips beneath a flat free surface, a boundary-normal extension uses one trace for the physical edge and its neighbouring Cartesian edge. Their common trace error cancels within the extension; a smooth blend connects it to the interior reconstruction. We give an error bound for the combined operator and test it against analytic cell means. In 21 initial-field cases, global velocity error is 2.71–32.07 times smaller than for a bilinear control using the same streamfunction. In subsequent smooth-wave calculations, the method also reduces the difference from same-grid numerical trajectories started with exact cell means. Against an independent linear two-fluid wave, it reduces endpoint physical surface-profile error by 2.81–5.49% on two grids when receiver velocity assignment is held fixed.

Keywords: conservative transfer; boundary integral method; streamfunction; cut cells; embedded boundaries; volume of fluid.

1 Introduction

Potential-flow boundary-integral-equation (BIE) methods represent strongly deformed free surfaces efficiently while the liquid remains irrotational and its boundary does not intersect itself. A two-phase field solver can subsequently represent flows outside that model class. Potential-to-field coupling is established: Lachaume et al. [1] combined boundary-element (BEM) and VOF methods for waves on slopes, and Colicchio et al. [2] developed BEM–level-set domain decomposition for strongly deformed and fragmented interfaces. More recent hybrid wave calculations include experimental comparisons and investigations of turbulence closures [3]. We focus on transferring a potential-flow state to a field solver.

Compatibility between velocity data and transport discretizations is a separate numerical problem [4]. Conservative remapping using geometric integration is well established [5], and streamfunction-based divergence-free interpolation has been developed for cut-cell fluid grids [6, 7]. Curl-Flow reconstructs a potential from staggered-grid velocities to obtain pointwise incompressible interpolation and enforce solid boundary conditions. Our input is instead a potential-flow boundary state, and the output of interest is the liquid-volume mean in each receiving cell. Taylor reconstruction combined with geometric moments also supports high-order embedded-boundary discretisations, including an unsteady Stokes method [8]. Here the physical trace and the adjacent Cartesian edge must be represented consistently in thin liquid strips. Table 1 compares the inputs, outputs, and target properties of these related constructions. We combine a BIE-derived liquid reconstruction with phase-volume moments on the geometry used by the receiver. This requires care with three distinctions: Cartesian face averages differ from liquid-cell means; Cartesian divergence cancellation does not guarantee zero embedded-volume flux divergence after loading; and independently clipped polygons may differ from the receiver’s solid geometry. The liquid-phase mean also differs from a density-weighted mixture velocity in a cell containing both phases; the independent two-fluid benchmark tests this receiver assignment separately.

Consistency of mass and momentum transport is especially important at large density contrasts. Earlier volume-tracking work [9] and subsequent staggered-grid VOF schemes [10] address this issue in the advection operator. Conservative VOF transport itself also depends on how the velocity field and directional stretching are represented [11]. Here we study the preceding initialisation step: a liquid moment and a mixture moment represent different quantities even when both are conservative. The density-weighted assignment used below is a standard projection; the factorial experiment measures its contribution separately from that of the new boundary extension.

Table 1: Related numerical constructions and the transfer problem considered here. The rows compare their inputs and objectives; they are not a ranking of accuracy.

Construction	Input and output	Main property	Relation to the present problem
Conservative remapping [5]	Discretised scalar fields on a source mesh to fields on a target mesh.	Global integral and constant fields; monotonicity for suitable maps.	Mesh-to-mesh transfer of scalar fields, rather than liquid velocity means from boundary data.
Curl-Flow [7]	Discretely divergence-free staggered velocities to a continuous velocity interpolant.	Pointwise incompressibility and solid-wall non-penetration.	Reconstructs a vector potential from grid velocities; the target is pointwise interpolation.
High-order embedded-boundary Stokes method [8]	Cell data and boundary conditions to a finite-volume time evolution.	High-order spatial operators with an approximate incompressibility projection.	Uses geometric moments and Taylor reconstruction to discretise a flow solver.
Present construction	Potential-flow boundary data to liquid-cell velocity means on receiver polygons.	Shared-edge liquid-moment balance; cancellation of common trace error in flat strips.	Matches the physical and adjacent Cartesian edge averages before division by strip thickness.

The general construction integrates liquid moments on receiver-matched polygons. Its internal edge contributions cancel by a standard discrete Green identity; this provides conservation on the represented geometry. The central accuracy improvement is narrower: for thin strips below a flat free surface, a boundary-normal extension uses the same trace on the physical and neighbouring Cartesian edges. This cancels the shared trace error before division by strip thickness. We test that mechanism against analytic cell means, propagated transfer error, and an independent physical wave. Additional tests of the general polygonal operator on an overturning state appear in Appendix A. The numerical scope is periodic, two-dimensional, square Cartesian grids, an impermeable single-valued bed, and a resolved non-self-intersecting free surface.

2 Transfer problem

Let Ω_ℓ be the liquid region in a periodic strip of length L . Its upper boundary Γ_f may overturn and its bed Γ_b is single-valued. The source supplies potential and normal-derivative traces for the harmonic problem

$$\Delta\phi = 0 \quad \text{in } \Omega_\ell, \quad \partial_n\phi = 0 \quad \text{on } \Gamma_b, \quad u = \nabla\phi. \quad (1)$$

The tested source has zero background current. We check boundary-flux compatibility before reconstructing the streamfunction and reject incompatible source states.

For a square Cartesian cell K of area h^2 , define

$$c_{s,K} = \frac{|K \cap \Omega_f|}{h^2}, \quad q_K = \frac{|K \cap \Omega_\ell|}{h^2}, \quad f_K = \frac{q_K}{c_{s,K}} \quad (c_{s,K} > 0), \quad (2)$$

where Ω_f is the non-solid domain. Thus $0 \leq q_K \leq c_{s,K} \leq 1$. The liquid mean velocity and its moment are

$$\bar{u}_{\ell,K} = \frac{1}{q_K h^2} \int_{K \cap \Omega_\ell} u \, dA, \quad m_K = q_K h^2 \bar{u}_{\ell,K}. \quad (3)$$

The density-normalised liquid momentum is $M_h = \sum_K m_K$. Every liquid-containing cell enters this sum, including small cut cells. The quantity $\bar{u}_{\ell,K}$ is distinct from both a whole-cell mean and a MAC face average. Initial momentum is a transfer target; it is not assumed constant during subsequent evolution under gravity and boundary forces.

3 Liquid streamfunction reconstruction

Cartesian elliptic solvers with irregular Dirichlet boundaries are well established; Gibou et al. [12], for example, give a symmetric second-order discretisation. We use the cut-distance scheme specified below. We use the convention

$$u = \partial_z \psi, \quad w = -\partial_x \psi, \quad \Delta\psi = 0. \quad (4)$$

On a boundary parameterised by $X(\alpha)$, the source normal velocity fixes the tangential derivative of ψ , with its sign determined by the boundary orientation. In counterclockwise liquid-boundary orientation, with arc length s and the outward normal, the convention in (4) gives $\partial_s \psi = \partial_n \phi$. The implementation uses periodic spectral integration on the upper boundary and a constant bed trace. The relative constant between the two boundary traces is fixed from the horizontal momentum identity of the source. Compatibility of the vertical momentum identities is checked independently. No velocity rescaling is applied afterward.

In particular, if both periodic boundary curves are traversed once with net horizontal displacement $+L$ and the bed trace is zero, then $M_x = \int_{\Gamma_f} \psi dx$. Adding a constant C to the free-surface trace changes this moment by CL , even for an overturning surface. The relative trace constant is therefore fixed by the source horizontal transport, whereas a common additive gauge on all boundaries leaves the velocity and closed-cell moments unchanged.

The harmonic Dirichlet problem is solved only at liquid grid vertices. If the distances to the left and right neighbouring values or boundary crossings are a_- and a_+ , the horizontal second derivative is approximated by

$$\frac{2}{a_- + a_+} \left(\frac{\psi_+ - \psi_0}{a_+} - \frac{\psi_0 - \psi_-}{a_-} \right). \quad (5)$$

The vertical expression is analogous. At a cut link, the prescribed physical trace replaces the exterior unknown. This cut-distance matrix is generally nonsymmetric; the implementation uses BiCGStab on CPU or restarted GMRES on GPU and checks the true residual after solving.

Exterior vertices required by liquid-containing cells receive a short normal Taylor continuation from the physical trace. The tested variant includes the quadratic term, with the normal second derivative obtained from harmonicity and boundary derivatives. A separate gas continuation then fills the remaining region while holding all liquid-support vertices fixed. The liquid solve and its moments are therefore unchanged by that gas completion. This isolation is an initialisation property; it does not remove the physical influence of gas during two-phase evolution.

4 Shared-edge liquid moments

4.1 Green identity and edge quadrature

Write $P_K = K \cap \Omega_{\ell,h}$ for the polygonal liquid portion of a cell. Orient each exterior ring counterclockwise and each hole clockwise. Applying Green's identity to (4) gives

$$m_K = - \oint_{\partial P_K} \psi (dx, dz). \quad (6)$$

For a directed straight edge from A to B , let $\Delta = B - A$, ψ_A, ψ_B be its endpoint values, and g_A, g_B their reconstructed gradients. The average of the cubic Hermite interpolant along the edge is

$$I_{AB} = \frac{\psi_A + \psi_B}{2} + \frac{(g_A - g_B) \cdot \Delta}{12}. \quad (7)$$

Its contribution to (6) is $-I_{AB}\Delta$. In the baseline operator, the physical-boundary trace is piecewise linear, so the correction in (7) is omitted. This keeps the physical trace integral identical to the one used to define the polygonal target. Cartesian edges use the Hermite rule in both full and cut cells. We call this *Linear* (L). *Hermite* (H) changes only the physical trace to cubic Hermite integration. *Matched* (M) also couples that trace to the neighbouring Cartesian edge in a flat surface strip through (10) and the blend below. *Bilinear* (B) takes the curl of a piecewise-bilinear interpolant of the same solved streamfunction and evaluates its liquid-polygon mean at the polygon centroid (the exact mean of that affine velocity field). It does not use the liquid edge integral. These names refer to transfer variants, not to the later receiver velocity assignment.

At Cartesian vertices the gradients are centred differences of the completed streamfunction, with second-order one-sided differences at artificial vertical endpoints. At a physical-boundary intersection in the baseline operator, the trace value and its two gradient components are linearly interpolated from the enclosing source-boundary segment. Thus the edge rule uses source derivatives at boundary intersections and grid derivatives at interior corners. The same endpoint data are reused by adjacent cells.

Proposition 1 (Discrete liquid-moment closure). *Suppose the liquid polygons form a conforming partition, every shared edge uses the same scalar integral with opposite orientation in its adjacent cells, and the periodic seam values match. Then the sum of the cell moments computed from (6) equals the integral over the exterior physical boundary of the polygonal liquid domain.*

Proof. Sum the oriented edge contributions over all polygons. Every internal edge appears twice with the same scalar integral and opposite directed edge vector, so the two contributions cancel. Periodic seam contributions cancel by the same argument. The remaining terms are precisely those on the physical boundary. In floating-point arithmetic the statement holds up to geometry and summation errors. $\square$

This identity concerns the polygonal reconstruction. Agreement with the continuous BIE momentum also depends on the boundary representation and source accuracy. The same edge rule must be used in full and cut cells for shared contributions to cancel.

4.2 Gauge subtraction and small-cell limitations

For any constant c_K , replacing ψ by $\psi - c_K$ in a closed-cell boundary integral leaves the moment unchanged because $\oint_{\partial P_K} (dx, dz) = 0$. The implementation uses a local constant to reduce cancellation when a small polygon is evaluated against a large streamfunction gauge.

Division by $q_K h^2$ in (3) is nevertheless a distinct source of sensitivity. For a fixed geometry, an absolute moment error δm_K induces a mean-velocity error $\delta \bar{u}_{\ell,K} = \delta m_K / (q_K h^2)$. Global closure therefore gives no uniform small-cell accuracy estimate. We report both phase-volume-weighted errors and errors by liquid fraction.

A direct perturbation bound makes this sensitivity explicit. Let δI_e be the error in the scalar edge average used in (6), on the same polygon as the reference, and let ℓ_e be its edge length. The triangle inequality yields

$$\|\delta \bar{u}_{\ell,K}\|_2 \leq \frac{\sum_{e \subset \partial P_K} \ell_e |\delta I_e|}{q_K h^2} \leq \frac{|\partial P_K|}{q_K h^2} \max_e |\delta I_e|. \quad (8)$$

For fixed geometry, this bound shows how edge-average errors grow when divided by a small liquid area. It does not bound the subsequent flow error.

The interpolation contribution can be isolated on the fixed polygonal geometry. Let $\tilde{\psi}(\sigma)$ be a prescribed twice continuously differentiable trace pulled back to each affected straight boundary chord, where σ is distance along that chord. Piecewise linear interpolation with maximum spacing $\delta \sigma$ satisfies $\|\tilde{\psi} - I\tilde{\psi}\|_\infty \leq (\delta \sigma)^2 \|\tilde{\psi}''\|_\infty / 8$. If this is the only perturbation and the affected physical edges of a cell have total length $P_{b,K}$, their contribution satisfies

$$\|\delta \bar{u}_{\ell,K}\|_2 \leq \frac{P_{b,K} (\delta \sigma)^2}{8 q_K h^2} \|\tilde{\psi}''\|_\infty. \quad (9)$$

The derivative is taken along the chord, rather than along the smooth BIE boundary. This bound excludes geometry error, propagation through the harmonic solve, and Cartesian-edge error. A fixed trace-to-grid spacing ratio therefore does not guarantee accuracy as the liquid fraction vanishes.

4.3 A consistent boundary extension for flat surface strips

Increasing the order of the physical trace alone need not remove the small-cell amplification: its integral and that on the neighbouring Cartesian edge can still use incompatible approximations. The Matched extension makes those two representations coincide as the strip thickness tends to zero. It is restricted here to a flat upper boundary.

Write $H(x)$ for a cubic Hermite trace on each physical boundary chord, with endpoint slopes supplied by the boundary streamfunction gradient. Let $N(x)$ and $S(x)$ be piecewise-linear interpolants of source-derived first and second normal derivatives. On a flat boundary the latter is obtained from harmonicity, $\psi_{zz} = -\psi_{xx}$, using differentiated source data. These quantities require no analytic interior solution. For a horizontal edge of width h a distance d below the surface, set

$$J(d) = \langle H \rangle - d \langle N \rangle + \frac{1}{2} d^2 \langle S \rangle, \quad \langle f \rangle = h^{-1} \int_{x_0}^{x_0+h} f(x) dx. \quad (10)$$

Every average is integrated over the restrictions of its parent chords. The physical upper edge uses precisely $\langle H \rangle$.

The implemented extension blends this edge average with the original Cartesian average I_C for $0 < d < h$:

$$I(d) = (1 - \beta(d/h))J(d) + \beta(d/h)I_C, \quad \beta(s) = 3s^2 - 2s^3. \quad (11)$$

It uses I_C for $d \geq h$. The two adjacent cells use the same blended edge integral.

Proposition 2 (Flat-strip error of the blended operator). *Let a rectangular liquid strip have width h , thickness $0 < d \leq h$, and flat upper boundary $z = z_s$. Suppose the exact streamfunction is C^3 in the closed strip and the physical and lower-edge approximations use the same $\langle H \rangle$ in (10). For the unblended edge $J(d)$ the horizontal mean is*

$$\bar{u}_J = \langle N \rangle - \frac{1}{2}d\langle S \rangle. \quad (12)$$

Define $E_N = \|N - \psi_z(\cdot, z_s)\|_\infty$, $E_S = \|S - \psi_{zz}(\cdot, z_s)\|_\infty$, and $C_3 = \|\psi_{zzz}\|_{\infty, P_K}$. The actual blended mean $\bar{u}_I = (\langle H \rangle - I(d))/d$ satisfies

$$|\bar{u}_I - \bar{u}| \leq E_N + \frac{1}{2}dE_S + \frac{1}{6}d^2C_3 + \frac{\beta(d/h)}{d}|I_C - J(d)|. \quad (13)$$

Proof. The horizontal Green moment is $h(\langle H \rangle - J(d))$. Division by hd gives (12); the same trace average cancels exactly. Taylor expansion of $\psi(x, z_s - d)$ at z_s , followed by horizontal averaging, bounds $|\bar{u}_J - \bar{u}|$ by the first three terms in (13). Equation (11) gives $\bar{u}_I - \bar{u}_J = \beta(d/h)(J(d) - I_C)/d$; the triangle inequality completes the proof. $\square$

Since $\beta(d/h)/d \leq 3d/h^2$, a bounded $|I_C - J(d)|$ produces no $1/d$ singularity as $d \rightarrow 0$ at fixed h . This alone does not prove uniform convergence under joint refinement. If N and S are piecewise linear interpolants of sampled normal derivatives with maximum horizontal spacing $\delta\sigma$, the indicated second horizontal derivatives are bounded, and the sample errors are at most η_N and η_S , respectively, then on each parent interval

$$E_N \leq \eta_N + \frac{(\delta\sigma)^2}{8}\|\partial_{xx}\psi_z(\cdot, z_s)\|_\infty, \quad E_S \leq \eta_S + \frac{(\delta\sigma)^2}{8}\|\partial_{xx}\psi_{zz}(\cdot, z_s)\|_\infty. \quad (14)$$

Here η_S includes errors from differentiating source data; it cannot be replaced by the error in the trace values. For example, suppose $\delta\sigma = O(h)$, $\eta_N = O(h^2)$, $\eta_S = O(h)$, the indicated derivatives and C_3 stay bounded, and $|I_C - J(d)| \leq Ch^3$ uniformly for $0 < d \leq h$. Then (13) gives a uniform $O(h^2)$ horizontal-mean bound on those strips. The final condition is a separate compatibility assumption on the Cartesian edge and the matched extension; it does not follow from an $O(h^2)$ pointwise streamfunction error alone. The estimate concerns the horizontal mean in flat strips. A vector-error bound for curved cut cells and floating-point effects at arbitrarily small volumes require separate analysis.

In a thin strip, the horizontal mean is the difference between two edge-averaged streamfunction values divided by d . Using a common trace preserves that small difference: its trace error cancels, whereas independent errors can dominate it (Figure 1).

4.4 MAC flux identity

Using streamfunction values at Cartesian vertices, the normal face averages can also be defined by endpoint differences. With consistent indexing,

$$U_{i,j+1/2} = \frac{\psi_{i,j+1} - \psi_{i,j}}{h}, \quad W_{i+1/2,j} = -\frac{\psi_{i+1,j} - \psi_{i,j}}{h}. \quad (15)$$

Their Cartesian cell divergence vanishes algebraically because the mixed differences commute. These face quantities are retained separately from the liquid moments. The identity does not include the receiver's open-face fractions, embedded boundary fluxes, or its interpolation on loading.

4.5 Optional projection to a mixture velocity

A one-fluid receiver stores one velocity per cell, whereas the preceding construction supplies a liquid-phase mean. If a gas-phase mean $\bar{u}_{g,K}$ is also prescribed, a density-weighted projection is

$$v_K^{\text{mix}} = \frac{\rho_\ell q_K \bar{u}_{\ell,K} + \rho_g(c_{s,K} - q_K) \bar{u}_{g,K}}{\rho_\ell q_K + \rho_g(c_{s,K} - q_K)} \quad (c_{s,K} > 0). \quad (16)$$

Multiplying by the cell fluid mass gives the sum of the represented phase momenta. Equivalently, this is the constant cell vector that minimises the density-weighted squared discrepancy from the two phase fields. This standard projection is distinct from assigning the liquid mean to every liquid-containing cell. In the tested surface strip with $q_K = 0.0005$ and $c_{s,K} = 1$, gas supplies about 70% of the cell mass, despite its small density; the two assignments can therefore differ appreciably. The phase-moment closure established above applies before this projection. The mixed vector must not subsequently be counted as a liquid-phase mean when checking that identity.

One trace. Two edge values.

Only their difference enters the thin-strip mean

- Reference edge values
- Approximated edge values

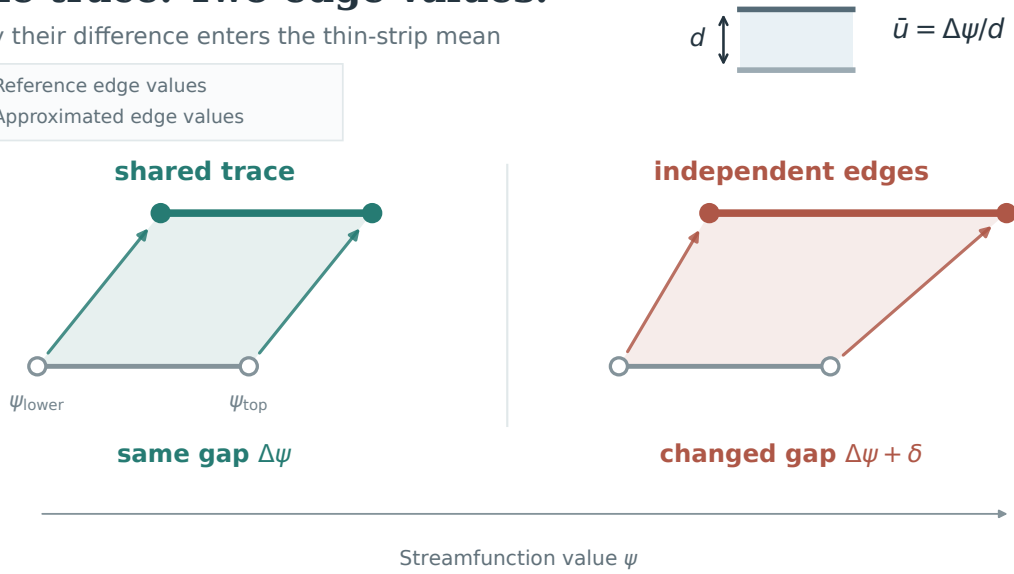


Figure 1: A shared trace preserves the difference of two edge values. The connected markers represent the two edge-averaged streamfunction values; horizontal position denotes value, while vertical separation distinguishes the reference and perturbed representations. Equal shifts translate the segment without changing its length; independent shifts deform it, introducing a mismatch divided by the small strip thickness d . The matched extension constructs both averages from the same physical trace, cancelling its common error before division. Errors in the normal derivatives remain, as quantified by (13). The diagram illustrates the unblended flat-strip construction; the Cartesian blend is described in the text.

We test this projection as an alternative receiver assignment, using a common prescribed gas phase for every transfer variant. Its usefulness depends on the receiver’s discrete variable and gas initialisation; the formula alone does not prove compatibility with an arbitrary VOF solver. The original native comparisons retain their liquid-mean assignment, and the independent two-fluid test compares the two assignments explicitly.

5 Matching the receiving geometry

The receiver is Aphros, an incompressible multiphase finite-volume solver with embedded boundaries and geometric VOF transport [13, 14]. Its embedded geometry is reconstructed from nodal signed-distance data. Preparing fractions from an independently clipped bed polygon can produce $q_K > c_{s,K}^{\text{native}}$. A loading operation that clips the liquid fraction to the available fluid volume then loses liquid even when the offline volume is exact.

The preparation procedure reproduces the receiver’s nodal edge crossings, constructs the corresponding bed, and integrates the liquid fractions and moments on that geometry. A small uniform surface-height adjustment reconciles the source volume with the polygonal volume; it is a geometric adjustment, not a redistribution of liquid among arbitrary cells. Its magnitude is measured for each prepared state. The embedded-bed reconstruction requires a resolved single-valued periodic graph and supported cell crossing topology; degenerate configurations are rejected.

After loading, the receiver applies its own pressure projection. Native embedded-volume flux divergence is measured independently of the offline MAC identity. The measurement protocol distinguishes input velocity, loaded velocity, and projected velocity; pressure-system residuals and embedded flux divergence are not interchangeable acceptance criteria.

The receiver uses the projection solver with AMGX for the symmetric pressure system. In the numerical units of the supplied state, the liquid and gas densities are 1 and 0.00117647, their dynamic viscosities are 2.5×10^{-5} and 4.88764×10^{-7} , surface tension is 0.001, and gravity is $(0, -1)$. The controls are CFL 0.05 and maximum time step 6.25×10^{-4} . These settings specify this initialisation experiment; the study does not calibrate them to a laboratory wave case.

Transfer procedure. The following steps specify how the geometry, streamfunction, and receiver data are coupled. Steps 1–3 and 5 apply to the general polygonal operator; step 4 is used only for the flat-strip Matched variant.

1. Reconstruct the receiver's embedded bed from its nodal signed-distance data, clip the liquid polygons on that geometry, and check source boundary flux compatibility.
2. Integrate the source normal-velocity trace to obtain the boundary streamfunction and fix the relative boundary constant from source transport. Solve the liquid harmonic Dirichlet problem and complete the vertices needed by the liquid polygons.
3. Give each shared polygon edge one scalar streamfunction integral, with opposite orientation in its two cells.
4. On flat upper strips, use the same cubic boundary trace on the physical edge and in $J(d)$; blend its neighbouring Cartesian edge through (11). Both adjacent cells use this shared edge integral.
5. Sum the oriented edge moments using (6), divide by liquid area, and supply the liquid means and matched fractions to the receiver. If the receiver instead requires a mixture velocity, use the separately defined projection (16) with prescribed gas means.

6 Numerical experiments

6.1 Error measures and test cases

For a reference liquid volume V_* and momentum M_* , report

$$e_V = \frac{|\sum_K q_K h^2 - V_*|}{V_*}, \quad e_M = \frac{\|M_h - M_*\|_2}{\|M_*\|_2}. \quad (17)$$

The relative momentum error is used only for a nonzero target. The analytic periodic tests have zero net momentum and retain absolute moment defects instead of dividing by their roundoff-sized target. For phase means on a selected set $\mathcal{S}$, an explicitly phase-volume-weighted relative norm can be defined by

$$e_{\ell,2}^2 = \frac{\sum_{K \in \mathcal{S}} q_K h^2 \|\bar{u}_{\ell,K} - \bar{u}_{*,K}\|_2^2}{\sum_{K \in \mathcal{S}} q_K h^2 \|\bar{u}_{*,K}\|_2^2}. \quad (18)$$

Probe and sampled-reference results below use their stated norms, which may differ from (18). Sources for the experimental code and data are given in [Data and code availability](#).

6.2 Native propagation of transfer error

A separate smooth-wave benchmark measures the subsequent influence of initial transfer error against a computable reference input. The periodic length is 64, the flat receiver bed is at $y_b = 0.53000001$, and the initially flat free surface is at $y_s = 1.5000625$. We use grids 512×32 , 1024×64 and 2048×128 , with the same physical geometry. Their upper liquid-row fractions are respectively 0.0005, 0.001 and 0.002. The prescribed initial potential is

$$\phi(x, y) = A \cos(kx) \frac{\cosh(k(y - y_b))}{\cosh(kD)}, \quad A = 0.02, \quad k = 2\pi 16/64, \quad D = y_s - y_b. \quad (19)$$

There is no background current. The reference input consists of exact liquid-cell means of $\nabla\phi$, integrated analytically over the rectangular liquid portions. An independent eight-point tensor Gaussian integration agrees with these means to 1.86×10^{-14} in relative liquid-weighted norm.

The principal comparison uses Matched (M), Bilinear (B) and exact initial means on one corrected receiver build. The boundary source has 512 nodes; trace factors 8, 8 and 16 give trace/grid-spacing ratios 0.125, 0.25 and 0.25. The first two levels reproduce the previous input policy, while the third retains the second level's ratio. Earlier Linear (L) and time-step comparisons are retained in Appendix A. Their geometry, gas velocities, physical coefficients and receiver settings are identical, and the receiver reconstructs its initial face fluxes from each branch's own cell velocities. Zero target momentum is checked with an absolute tolerance 10^{-6} , rather than an ill-defined relative error. The physical coefficients are those listed above. The nominal inviscid capillary-gravity time scale, neglecting gas density in the dispersion relation [15], is $T = 2\pi / \sqrt{(gk + \sigma k^3 / \rho_\ell) \tanh(kD)} = 5.25072$; it is used only to specify the common endpoint $t = 0.1T = 0.52507177$. The reference branch is a numerical continuation from exact initial means, not an exact later-time two-phase solution.

With R denoting the exact-input reference and A a tested initialisation, the primary global error is

$$E_u^2(t) = \frac{\sum_K q_K^R(t) \|u_K^A(t) - u_K^R(t)\|_2^2}{\sum_K q_K^R(t) \|u_K^R(t)\|_2^2}. \quad (20)$$

VOF and column-depth differences use the reference branch in their relative norm denominators. The primary runs use $\Delta t_{\max} = 0.0025$ and CFL 0.1. The earlier half-step control uses 0.00125 and CFL 0.05.

7 Results

The main comparisons concern the Matched extension on flat surface strips. Tests of the general polygonal construction and earlier transfer variants are reported in Appendix A.

7.1 Analytic initial-field tests of the Matched extension

We evaluate the extension in (10) in 15 cases: mode-4 manufactured fields at $n_x = 128, 256$, with upper-strip liquid fractions $10^{-5}, 10^{-3}, 10^{-2}, 10^{-1}$; mode-16 wave fields at $n_x = 512$ with fractions $0.0005, 0.001, 0.01, 0.1$; and mode-16 fields at $n_x = 1024$ with fractions $0.001, 0.01, 0.1$. Closed-form phase-cell means provide the reference. Source data, geometry and the comparison cases are fixed within each comparison. No analytic interior field is used by the matched operator.

Hermite (H), which changes only the physical trace, isolates the role of edge compatibility. In the manufactured 128-column case at $q_{\text{top}} = 0.001$, its top-row relative error rises to 0.08046 from 0.03757 for Linear (L). Matched (M) lowers it to 9.07×10^{-6} . For the wave case with 512 columns and $q_{\text{top}} = 0.0005$, the three L, H, and M errors are 0.18570, 0.15833 and 3.70×10^{-5} . This ablation supports a major role for the mismatch of edge representations: increasing the physical-trace order alone does not deliver the observed improvement.

M reduces both the top-row error and the all-liquid, volume-weighted error in all 15 cases relative to L and H. It also improves the global error relative to B in all 15 cases, by factors 2.71–12.60 (Table 2). Its maximum top-row relative error over this matrix is 5.16×10^{-5} ; the maximum absolute moment-closure defect is 2.61×10^{-15} . For the original wave geometries, the global initial error decreases from 0.002987 to 0.001420 at 512 columns and from 0.004700 to 0.0003538 at 1024 columns. The remaining global error includes the interior and bed cells, which the surface extension does not directly correct. These measurements concern initial phase means; the evolved-field comparisons follow below.

Table 2: Initial velocity errors for the Matched extension in the liquid-volume-weighted relative norm over all liquid cells. F4 is the manufactured mode-4 field; W16 is the mode-16 wave field. Bilinear uses the same computed streamfunction. Gain is the Bilinear-to-Matched error ratio.

Family	n_x	q_{top}	Linear (L)	Matched (M)	Bilinear (B)	Gain
F4	128	1e-05	7.049e-03	1.475e-03	3.993e-03	2.71
F4	128	0.001	1.678e-03	1.474e-03	4.184e-03	2.84
F4	128	0.01	1.553e-03	1.462e-03	5.606e-03	3.83
F4	128	0.1	1.603e-03	1.356e-03	1.202e-02	8.87
F4	256	1e-05	1.693e-03	3.526e-04	9.936e-04	2.82
F4	256	0.001	3.870e-04	3.524e-04	1.092e-03	3.10
F4	256	0.01	3.548e-04	3.507e-04	1.738e-03	4.96
F4	256	0.1	3.614e-04	3.352e-04	4.223e-03	12.60
W16	512	0.0005	2.987e-03	1.420e-03	4.087e-03	2.88
W16	512	0.001	2.324e-03	1.420e-03	4.183e-03	2.95
W16	512	0.01	1.503e-03	1.407e-03	5.612e-03	3.99
W16	512	0.1	1.470e-03	1.298e-03	1.205e-02	9.28
W16	1024	0.001	4.700e-03	3.538e-04	1.090e-03	3.08
W16	1024	0.01	1.495e-03	3.520e-04	1.735e-03	4.93
W16	1024	0.1	5.098e-04	3.361e-04	4.218e-03	12.55

A further six cases use mode 7 at $n_x = 512, 1024$ and $q_{\text{top}} = 0.3, 0.7, 0.99$ with the same operator. The global error remains lower than B in all six cases, by factors 3.73–32.07. These larger fractions also exercise the transition back to Cartesian integration. At $q_{\text{top}} = 0.99$ the M and H top-row errors nearly coincide, with the M value about 0.2% larger; the matched extension therefore is not uniformly superior to that ablation at all fractions. Its moment-closure defect remains below 5.32×10^{-17} .

Table 3 tests the final M operator on three grids while refining the boundary trace with the receiver: its spacing is $h/8$ throughout. The global velocity error decreases from 1.4203×10^{-3} to 3.5378×10^{-4} and 8.0414×10^{-5} , with observed orders 2.005 and 2.137. B uses the same computed streamfunction and has larger errors on all three grids. These are transfer-refinement results for a fixed boundary source state; they do not establish convergence of the upstream boundary solver. The cut-cell errors and maximum vector errors are retained with the global norms in the accompanying data.

Table 3: Initial liquid-volume-weighted velocity error under joint grid and trace refinement (W16). The refined trace spacing is $h/8$ on each grid. Orders compare successive rows; the source boundary state has 512 nodes throughout.

n_x	Trace points	E_M	Order	E_B	Order	E_B/E_M
512	4096	1.4203e-03	—	4.0869e-03	—	2.88
1024	8192	3.5378e-04	2.005	1.0898e-03	1.907	3.08
2048	16384	8.0414e-05	2.137	3.3498e-04	1.702	4.17

Table 4: Propagated transfer error at $t = 0.1T$ on the common corrected receiver build. Each reference trajectory starts from exact cell means on the same grid. E_u is the liquid-weighted velocity error and E_H the wave-normalised column-depth discrepancy; $R = E_B/E_M$.

n_x	E_u^M	E_u^B	R_u	E_H^M	E_H^B	R_H
512	1.7585e-03	1.3861e-02	7.88	3.9263e-05	2.5528e-03	65.02
1024	4.3460e-04	4.2597e-03	9.80	7.3386e-06	5.4177e-04	73.83
2048	9.8215e-05	1.7489e-03	17.81	1.6265e-06	1.6651e-04	102.38

7.2 Propagation of transfer error

The three-grid comparison uses the same corrected receiver executable and library for all nine M, B and exact-input branches. Common geometry, gas velocities, physical coefficients and time-step controls isolate the initial transfer. The receiver reconstructs its face fluxes from each branch’s cell velocities; the common supplied face arrays are not imposed as runtime fluxes.

Table 4 reports the errors at $t = 0.1T$. M reduces the global velocity discrepancy by factors of 7.88, 9.80 and 17.81 on the three grids. The corresponding surface-profile reductions are 65.0, 73.8 and 102.4. Both improvements hold at every recorded nonzero time (Fig. 2). The maximum recorded relative liquid-mass discrepancy is 7.15×10^{-13} , and the maximum post-start flux divergence is 6.99×10^{-14} .

The M endpoint velocity errors are 1.7585×10^{-3} , 4.3460×10^{-4} and 9.8215×10^{-5} . Their two-level observed orders are 2.017 and 2.146. This native series uses the trace policy specified in Section 6.2; the fixed trace/grid-ratio initial test is reported separately in Table 3. The earlier full- and half-step comparisons are retained in Appendix A.11; the three-grid results here establish the transfer advantage on one documented corrected receiver build.

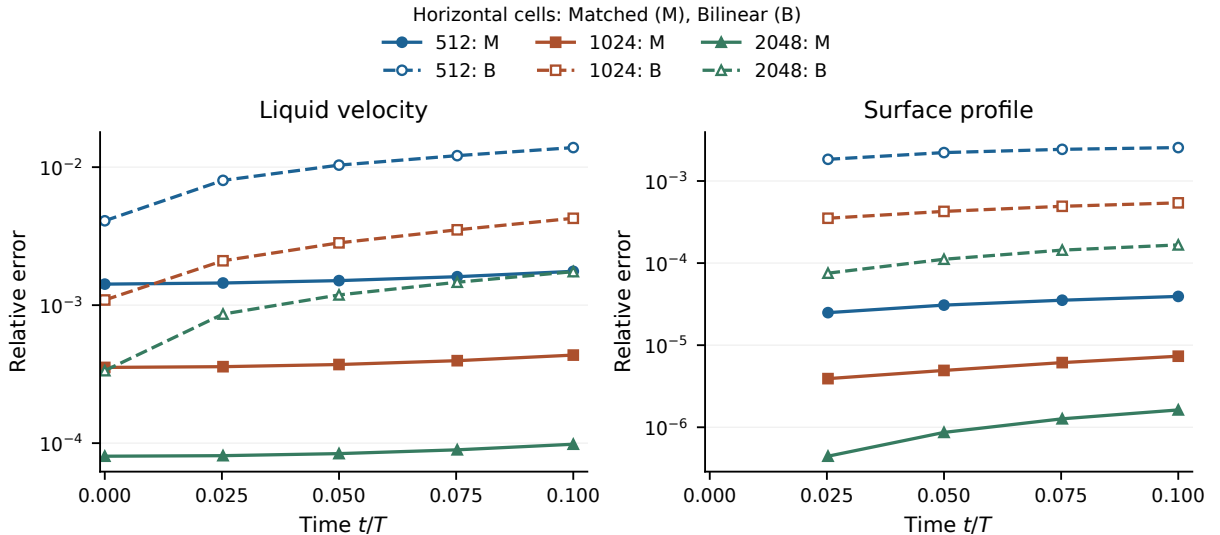


Figure 2: Propagated transfer error on three grids using the common corrected receiver build and $\Delta t_{\max} = 0.0025$. Each curve is compared with the exact-input numerical trajectory on its own grid. Filled markers and solid lines denote M; open markers and dashed lines denote B. Surface error is shown at every nonzero output time; its normalisation is undefined for the initially flat profile.

7.3 Independent small-amplitude two-fluid wave

An additional benchmark compares the evolved column depth with an independent linear two-fluid capillary-gravity wave. It uses the same physical geometry and density ratio, but zero viscosity, a slip bottom and a pressure-release upper boundary. The assignment comparison uses 512×32 cells; its spatial check adds 1024×64 cells. Let $D_\ell = y_s - y_b$ and $D_g = 4 - y_s$. The linear frequency and surface amplitude for initial liquid potential $A \cos(kx) \cosh(k(y - y_b)) / \cosh(kD_\ell)$ are

$$\omega^2 = \frac{(\rho_\ell - \rho_g)gk + \sigma k^3}{\rho_\ell \coth(kD_\ell) + \rho_g \tanh(kD_g)}, \quad B = \frac{Ak \tanh(kD_\ell)}{\omega}. \quad (21)$$

The corresponding initial gas potential is $-A \tanh(kD_\ell) \sinh(k(4 - y)) \cos(kx) / \cosh(kD_g)$. It satisfies the linear kinematic condition at the interface and the pressure-release condition at the top. We check the boundary-equation residuals and evaluate the phase-cell means independently by quadrature. The reference column depth averaged over a horizontal cell is

$$H_i^{\text{lin}}(t) = D_\ell + B \sin(\omega t) \text{sinc}(kh/2) \cos(kx_i), \quad \text{sinc}(s) = \sin(s)/s. \quad (22)$$

Here the depth-error RMS is normalised by the fixed quantity $B \text{sinc}(kh/2) / \sqrt{2}$; this differs from the instantaneous numerical-reference normalisation in Fig. 2. The endpoint is $t = 0.1(2\pi/\omega) = 0.52566088$ and $\Delta t_{\text{max}} = 0.00125$.

A zero-amplitude control exposed an undefined zero-displacement operation in the receiver's two-dimensional geometric VOF flux routine. Returning zero transported volume for zero displacement prevents the 0.403% first-step liquid loss of the unmodified build. The corrected rest control preserves mass and has endpoint depth-error RMS 1.11×10^{-16} . The physical-reference tests and the three-grid propagation comparison use this same corrected build. The earlier unmodified-runtime comparisons are retained in Appendix A.11.

With exact initial phase means, $A = 2 \times 10^{-5}$ gives normalised depth error 0.03533528 at the endpoint; $A = 10^{-5}$ gives 0.03533601. The absolute errors are 5.96216×10^{-7} and 2.98114×10^{-7} . Their near-linear amplitude scaling and the separate rest control support comparison with the linear reference at these amplitudes. This reference tests a smooth inviscid wave, not a nonlinear breaking-wave solution.

For the transfer comparison, M and B liquid inputs are scaled by 10^{-3} to $A = 2 \times 10^{-5}$, without retuning the operator. The corresponding scaled exact liquid field agrees with the independently integrated phase means within 5.88×10^{-17} in maximum absolute velocity. Geometry and dry/gas cell velocities are identical across the three branches, and each branch uses the same native flux-reconstruction procedure. Table 5 reports the complete nonzero-time comparison. At the endpoint, M reduces the normalised physical depth error from 0.03718424 to 0.03535843, a 4.91% reduction. It improves this metric at every recorded nonzero time, with reductions from 4.91% to 13.01% over the four samples.

The discrepancy from the exact-input numerical trajectory is much smaller: its normalised depth RMS is 2.31536×10^{-5} for M and 1.84896×10^{-3} for B, a factor of 79.86. Their global liquid-weighted velocity discrepancies are 0.00168245 and 0.01070309, a factor of 6.36. The moderate 4.91% improvement against the analytic wave is therefore compatible with a much larger reduction of the transfer contribution: the exact-input receiver already has a physical depth error of 0.03533528. Each of these three branches preserves liquid mass; the maximum measured post-start flux divergence is below 10^{-15} . The three branches share endpoint times, geometry, and receiver settings.

Table 5: Depth-error RMS against the independent linear two-fluid wave, normalised by the fixed analytic wave-amplitude RMS. All recorded nonzero times are included. The last column is the relative reduction from Bilinear (B) to Matched (M) transfer.

t	Exact input	Matched (M)	Bilinear (B)	Reduction (%)
0.13125	0.00248271	0.00248760	0.00285971	13.01
0.26250	0.00811344	0.00812471	0.00900079	9.73
0.39375	0.01821401	0.01823153	0.01960601	7.01
0.52566	0.03533528	0.03535843	0.03718424	4.91

The alternative assignment in (16) uses the same wave protocol. With exact phase means it reduces the endpoint physical depth error from 0.03533528 to 0.01775465. At half amplitude the corresponding error is 0.01775438, again preserving the linear scaling, and both runs preserve liquid mass. This experiment supports the density-weighted assignment for this receiver and state; it does not establish a universal VOF velocity contract.

The complete factorial comparison retains all six combinations of phase reconstruction and receiver assignment (Table 6). With density-weighted projection, M gives endpoint physical depth error 0.01777826, compared

with 0.01881090 for B: a 5.49% reduction under identical receiver assignment. The combined M-plus-mixture construction reduces the error from the original B liquid-mean assignment, 0.03718424, by 52.19% (a factor of 2.092). The latter improvement includes both changes and must not be attributed to the boundary extension alone.

The combined construction improves on the original B assignment at every recorded nonzero time. Within the mixture assignment, however, the B control has the smaller physical error at the first sample ($t = 0.13125$: 0.00016171 versus 0.00040606); M is better at the three later samples. Lower initial transfer error therefore need not minimise total physical error at each individual time. Against the exact-mixture-input numerical trajectory, the endpoint depth discrepancies are 2.36173×10^{-5} and 1.05626×10^{-3} for M and B, respectively. These transfer-only discrepancies are distinct from the physical errors in the table. The mixed-cell inputs use separately integrated gas-phase means and satisfy the density-weighted formula to roundoff.

Table 6: Factorial comparison at 0.17 against the independent linear two-fluid wave: normalised physical depth-error RMS. Each column uses the same phase reconstruction under both receiver assignments; geometry, gas phase and runtime are common.

Receiver assignment	Exact phase data	Matched (M)	Bilinear (B)
Liquid mean in wet cells	0.03533528	0.03535843	0.03718424
Density-weighted mixture	0.01775465	0.01777826	0.01881090

The spatial check repeats M, B and exact-input runs with mixture assignment on 512×32 and 1024×64 cells, holding the amplitude, physical geometry, boundary conditions and time-step limit fixed. Table 7 includes every recorded nonzero time. The coarse-grid endpoint reproduces the 5.49% reduction. On the finer grid, M reduces physical depth error from 0.00985114 to 0.00957410, a 2.81% reduction, and improves on B at all four nonzero samples. The M endpoint error decreases by a factor of 1.857 under refinement; the exact-input error decreases from 0.01775465 to 0.00957065. The remaining physical error is therefore close to that of the exact-input receiver, even though the transfer contribution is much smaller with M. All six runs preserve liquid mass at output precision, with maximum recorded post-start flux divergence 5.51×10^{-15} .

7.4 Computational cost of the transfer

Table 8 isolates the in-memory Matched transfer on an Intel Core Ultra 7 155H CPU under Windows 11, using Python 3.13.3, NumPy 2.3.5, SciPy 1.17.1 and Shapely 2.1.2. The trace factors are 8 and 16 on the two grids. Each timing uses three repetitions after one warm-up; the output velocity arrays are byte-identical across repetitions. The measured transfer includes geometry, the liquid solve, gas continuation, cell moments and flux preparation, but excludes serialization, checkpoints and subsequent flow evolution. Boundary-trace preparation is timed separately. Liquid-cell moment construction accounts for approximately half of the median transfer time in this implementation. These measurements describe the current CPU code, rather than a GPU speedup or an asymptotic complexity estimate.

Table 7: Physical column-depth error against the cell-averaged linear two-fluid wave, with density-weighted receiver assignment held fixed. Errors use the fixed linear-wave RMS amplitude. Every recorded nonzero time is included; the reduction is $100(1 - E_M/E_B)$.

n_x	t	Exact input	Matched (M)	Bilinear (B)	Reduction (%)
512	0.13125	0.00041102	0.00040606	0.00016171	-151.10
512	0.26250	0.00026377	0.00027524	0.00079983	65.59
512	0.39375	0.00529807	0.00531594	0.00610920	12.98
512	0.52566	0.01775465	0.01777826	0.01881090	5.49
1024	0.13125	0.00006724	0.00006795	0.00012428	45.33
1024	0.26250	0.00075300	0.00075460	0.00088189	14.43
1024	0.39375	0.00337341	0.00337593	0.00357862	5.66
1024	0.52566	0.00957065	0.00957410	0.00985114	2.81

Table 8: CPU transfer cost in seconds: median [minimum, maximum] over three repetitions after one warm-up. Stages exclude file output, checkpoints and CFD evolution. Stage medians need not sum to the median total.

Stage	512×32	1024×64
Boundary trace	0.0212 [0.0167, 0.0225]	0.0450 [0.0409, 0.0483]
Geometry and harmonic-system assembly	0.1028 [0.1020, 0.1114]	0.4387 [0.3556, 0.4480]
Liquid harmonic solve	0.0018 [0.0017, 0.0018]	0.1013 [0.0539, 0.1332]
Gas continuation	0.0144 [0.0085, 0.0192]	0.3019 [0.2840, 0.3689]
Liquid-cell moments	1.9355 [1.8201, 2.2680]	12.3647 [10.1513, 12.4287]
Remaining in-memory operations	1.8810 [1.7302, 3.2785]	11.1201 [10.4661, 12.9108]
Transfer total, excluding trace	3.8285 [3.7779, 5.6702]	23.5884 [22.1545, 26.1844]

8 Discussion and limitations

The main gain comes from coupling two edge representations. Raising the physical-trace order alone does not remove their mismatch in a thin strip. The bound (13) explains cancellation of the common trace error, while the derivative errors and the Cartesian blend remain. It gives a uniform refinement rate only under the stated bounds on derivative sampling and $I_C - J(d)$; those bounds have not been proved for the coupled harmonic solve. The observed refinement rates therefore complement the local mechanism rather than establish an unconditional convergence theorem.

The smooth-wave trajectories started from exact cell means isolate the effect of initial transfer under a common numerical receiver. Their error against each other should not be read as physical error. The independent linear two-fluid wave supplies a physical reference and shows a smaller gain. Receiver velocity assignment also matters: the 52.19% combined reduction on the coarse grid includes both Matched transfer and the standard mixture projection, whereas the 2.81–5.49% comparisons hold assignment fixed. With mixture assignment, B is better at the first recorded nonzero time on the coarse grid; M is better at the three later samples and all four fine-grid samples. Cross-step comparisons can reverse the ranking because temporal and transfer errors interact (Appendix A).

The polygonal moment construction has a wider geometric scope than the flat-strip extension, but its discrete closure does not ensure local accuracy. The earlier Linear variant improves selected native bed-cut cells over B, while an exact-field centroid sample is more accurate in the nine pooled flat cut-cell tests (Appendix A). The fixed overturning state has surface-height span 0.77104 and a non-monotone horizontal parametrisation. M rejects this geometry; its transfer and native startup test only the general polygonal construction. The fixed-state continuation has no independent evolved reference.

The tests use two-dimensional periodic geometry and zero background current. The local estimate covers the horizontal mean in flat strips; the independent BIE sample covers 12 selected bed-cut cells. The smooth native runs are short, and neither the upstream wave formation nor a complete breaking-wave trajectory has been shown to converge in space and time. An extension to overturning boundaries would require a curved-boundary construction and its own accuracy analysis.

9 Conclusions

For thin strips beneath a flat surface, Matched transfer uses one physical trace on both sides of the edge difference. The common trace error cancels before division by the small strip thickness. The bound for the implemented blend identifies the remaining derivative and Cartesian-edge errors. Across 21 analytic initial-field tests, M lowers global velocity error relative to B. A separate three-grid initial test gives observed orders 2.005 and 2.137 under joint trace/grid refinement. On a common corrected receiver build, the three-grid continuations reduce propagated velocity error by factors of 7.88–17.81 and surface-profile error by 65.0–102.4 against same-grid numerical trajectories started from exact cell means.

Against an independent linear two-fluid wave, M lowers endpoint physical surface-profile error by 5.49% and 2.81% on the two grids when mixture assignment is held fixed. On the coarse grid, combining M with that standard assignment lowers the error by 52.19% relative to the original practical B input with liquid-mean assignment; this larger reduction includes both changes. The more general polygonal construction preserves discrete liquid moments by shared-edge cancellation, and receiver-matched geometry avoids liquid loss on loading. The flat-strip accuracy estimate does not apply to overturning surfaces, and these short continuations do not validate a complete breaking-wave trajectory.

Data and code availability

The original code, benchmark inputs and numerical evidence are archived as v0.31.0 [16]. The additional refinement, dynamic, physical-reference and cost experiments are archived in v0.32.0 [17]. It includes the input and output fields, analysis scripts, transfer sources, corrected receiver source and patch, build records, and runtime configuration hashes used in the principal comparisons. Its verifier recomputes the three main numerical audits from the archived fields. Historical source-provenance limits remain documented with the original archive; its DOI does not identify the later experiments.

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Additional information

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A General polygonal transfer and earlier variants

These tests document the base polygonal operator, trace resolution, and receiver sensitivity. They do not use the flat-strip Matched extension on the overturning geometry.

The fixed wave state has 2304 nodes on each boundary and comes from a refined boundary solve on a fixed geometry. The convergence of its upstream nonlinear BIE trajectory was not assessed here. Its target liquid volume is 40.52627859944759. The grids are 1024×64 , 2048×128 , and 4096×256 for $L = 64$. The finest spacing is $h = 0.015625$. Refining these receiver grids with the same source tests the transfer and receiver sensitivity, not the convergence of the upstream time-dependent BIE solution. Figure 3 shows the source boundary geometry.

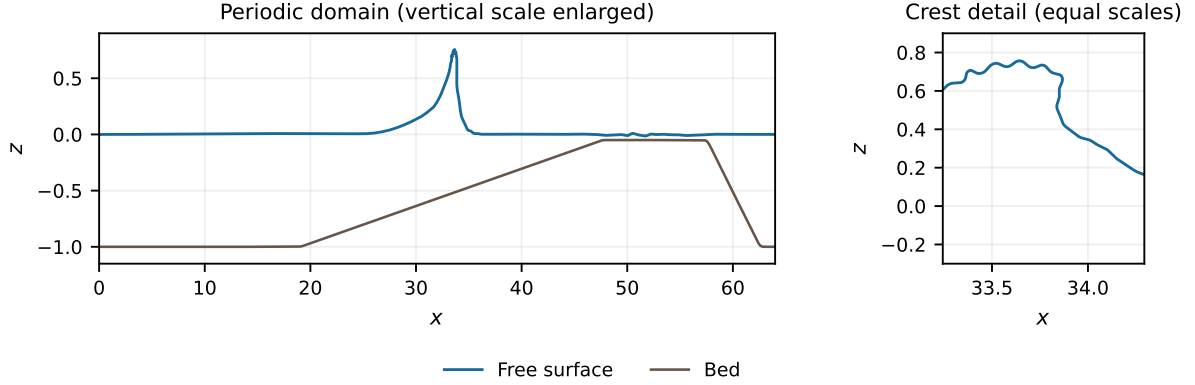


Figure 3: The fixed wave boundary and bed used in the offline transfer tests, before matching the receiver’s bed convention. The curves are drawn directly from the source polygon without smoothing. The left panel enlarges the vertical scale; the crest detail uses equal scales. Small surface oscillations are part of this supplied state, whose upstream trajectory convergence is not claimed.

A.1 Analytic refinement and controlled comparisons

The manufactured test has flat boundaries $z = -D$ and $z = 0$, with $D = 4$, $L = 64$, $k = 2\pi 4/L$, and

$$\phi(x, z) = \frac{\cos(kx) \cosh(k(z + D))}{\cosh(kD)}, \quad \psi(x, z) = -\frac{\sin(kx) \sinh(k(z + D))}{\cosh(kD)}. \quad (23)$$

The vertex grid starts at $z = -D - 0.37h$ to produce cut rows. The reference is obtained by analytic cell-volume integration over the clipped rectangles. The reported norm is the unweighted Euclidean norm over the liquid-cell mean components, divided by the corresponding reference norm. This avoids confusing a cell mean with a point value at the cell centre. Boundary trace representation is refined together with the Cartesian mesh; the source uses 256 points per boundary and factors 8, 16, and 32 yield 2048, 4096, and 8192 trace points. The flat boundary geometry itself is exact. A second test holds the trace sampling factor fixed to expose the resulting limitation.

A.2 Native comparison protocol

The practical control uses the curl of the piecewise-bilinear interpolant of the same solved Cartesian streamfunction. In a cell with local coordinates $\xi = (x - x_i)/h$, $\eta = (z - z_j)/h$ and corner values $p_{00}, p_{10}, p_{01}, p_{11}$, write $d = p_{11} - p_{10} - p_{01} + p_{00}$. Its velocity is

$$u = \frac{p_{01} - p_{00} + d\xi}{h}, \quad w = -\frac{p_{10} - p_{00} + d\eta}{h}. \quad (24)$$

Each component is affine, so its exact mean over a liquid polygon equals its value at that polygon’s area centroid. This is a computable liquid-domain control, distinct from both whole-cell arithmetic averaging and the exact-field centroid control used in the flat test.

The control and L initialisations use identical liquid fractions, embedded geometry, gas velocities, source streamfunction, receiver executable, and solver parameters. The native receiver reconstructs its initial face fluxes from each input cell-velocity field and applies the same projection. No momentum correction is applied to the control. Its initial momentum error, 1.100539×10^{-6} , exceeds the 10^{-6} tolerance used for the L input. The control

comparison therefore uses a 2×10^{-6} momentum tolerance, while mass, finite-field, flux-divergence and runtime checks are unchanged.

Comparisons use the exact common final time $t = 0.01$. Scheduled intermediate dumps differ in time by up to 9.24×10^{-6} between the two branches and are not treated as simultaneous fields. With superscripts L and B denoting Linear and Bilinear solutions, define

$$D_u^2 = \frac{\sum_K q_K^L \|u_K^L - u_K^B\|_2^2}{\sum_K q_K^L \|u_K^L\|_2^2}, \quad D_f = \frac{\|f^L - f^B\|_2}{\|f^L\|_2}. \quad (25)$$

Column liquid depths are $H_i = h \sum_j q_{ij}$, with relative difference $D_H = \|H^L - H^B\|_2 / \|H^L\|_2$. These quantities measure sensitivity to the initial reconstruction; neither branch is a physical reference.

A single additional intervention increases the boundary-trace factor from 8 to 32 for both reconstructions, keeping the 512 source nodes and physical geometry fixed. Both trace densities are compared. The exact-input reference, liquid fractions, embedded geometry, and gas velocities are held fixed in this control. Recomputed preprocessing face velocities change only at numerical-noise scale; the serialized common face arrays are copied unchanged from the original reference. Native face fluxes are reconstructed separately from each branch's cell velocities. For this flat surface, the streamfunction trace is $-A \tanh(kD) \sin(kx)$, with maximum second derivative 0.04487475. Its original and refined spacings are $1/64$ and $1/256$. For the upper liquid-strip height $q_K h = 6.25 \times 10^{-5}$, equation (9) bounds the isolated physical-trace contribution by 0.02191 and 0.001369 respectively. These are upper bounds on one error component, not estimates of the total transfer or evolved error.

A final joint-refinement check uses 2048×128 cells and trace factor 64, preserving $\delta\sigma/h = 1/16$ from the denser-trace 1024×64 case. The 512 source nodes, geometry, potential amplitude, wave mode and endpoint remain fixed; the upper-row liquid fraction becomes 0.002. This case uses only the full time-step limit. The time-step halving coverage consists of both original-trace grids and the denser-trace 1024×64 case; no half-step trajectory is inferred for untested cases.

A.3 Manufactured field

Table 9 shows matched grid and trace refinement. The observed order is $p = \log_2(e_h/e_{h/2})$ for this flat-boundary test.

Table 9: Joint refinement of Cartesian resolution and boundary-trace sampling on exact flat boundaries.

Horizontal cells	Boundary factor	Reported relative error	Observed order
128	8	$1.7828103751 \times 10^{-3}$	—
256	16	$4.0059606173 \times 10^{-4}$	2.154
512	32	$9.1208099630 \times 10^{-5}$	2.135

With the boundary representation held at factor 8, the final observed order falls to about 1.505, showing sensitivity to trace resolution even on flat geometry. CPU and GPU results agree: at L10 the largest velocity difference is 2.37×10^{-10} , with identical liquid liquid fractions.

A.4 Small liquid fractions and an analytic sampling control

A nine-case stress test retains the harmonic potential above and places the free surface at $z_s = (-0.37 + q_s)h$, so that the top liquid row has a prescribed fraction $q_s \in \{0.001, 0.01, 0.1\}$. The bed remains at $-D$. This is a family of slightly shifted flat domains, each compared with its own exact phase-volume means. Boundary potentials and normal derivatives are evaluated consistently at the shifted surface. Horizontal resolutions are 128, 256, and 512 with the same joint trace-factor policy as above.

Three reconstructions are compared before any native receiver projection: the L shared-edge moments; the arithmetic average of neighbouring MAC face values from the same solved streamfunction; and the exact harmonic velocity sampled at the centroid of the liquid rectangle. The last uses the exact interior field and is a sampling control rather than a BIE evaluation algorithm. The reference for all three is the analytic liquid-volume mean. The reported cut-cell norm includes both the surface and bed cut rows, without phase-volume weights.

Figure 4 shows how cut-cell accuracy depends on liquid fraction. For $q_s = 0.001$, the L cut-cell errors are 0.03585, 0.01057, and 0.008982. For $q_s = 0.01$, they are 0.006350, 0.0005889, and 0.0007439, including a nonmonotone final step. At $q_s = 0.1$, the errors decrease from 0.003427 to 0.0007165 to 0.0001358. The analytic

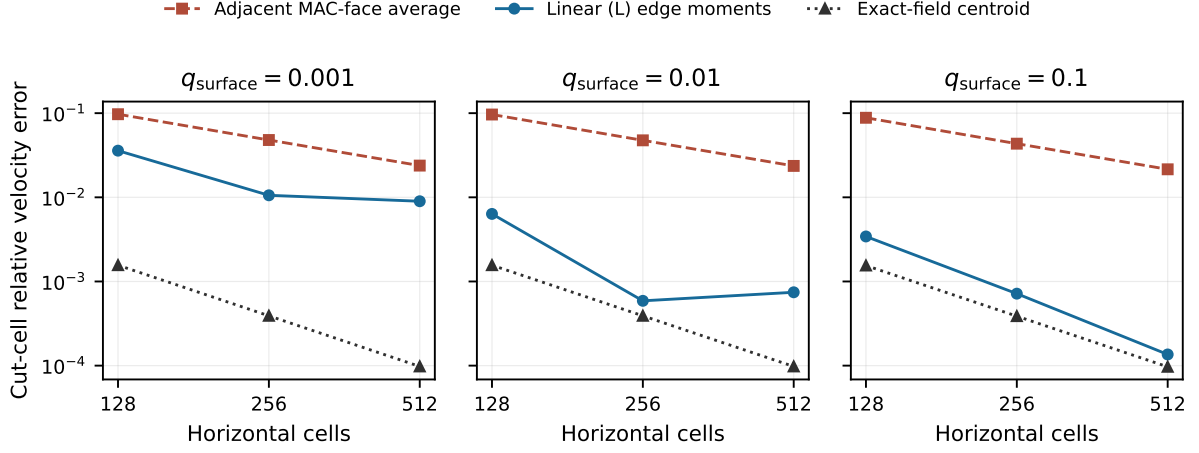


Figure 4: Flat-boundary stress tests before native projection. The prescribed surface fraction is shown above each panel; the error norm includes both cut rows. The L reconstruction improves on adjacent-face averaging, but the exact-field centroid control is more accurate in all nine cases.

centroid control is more accurate throughout, while simple MAC-face averaging is less accurate. Liquid volume remains conserved to roundoff in all cases. Thus conservation and improvement over arithmetic averaging do not imply lower error than sampling the exact liquid-domain field, or uniform second-order behaviour in thin liquid regions.

To isolate trace-sampling sensitivity, two additional runs increase the number of boundary trace samples fourfold while holding the Cartesian grid and the analytic domain fixed, with $q_s = 0.001$. At 256 horizontal cells, the cut-cell error decreases from 0.0105686 to 0.0087533; at 512 cells, it decreases from 0.0089818 to 0.0007110. These are reductions by factors 1.21 and 12.63, respectively. The geometry remains exactly flat, and the exact-field centroid control is unchanged. Thus trace resolution is a material source of error, but this control does not establish that it is the only source or that additional sampling yields uniform convergence as $q_s \rightarrow 0$.

A comparison over all liquid cells uses the liquid-volume-weighted norm in equation (18), rather than the unweighted cut-cell norm in Figure 4. Across the same nine cases, the L reconstruction reduces this global error by factors 4.30–27.05 relative to arithmetic MAC averaging. However, its global errors are 4.61–27.39 times larger than those of the exact-field centroid control. Thus a global accuracy advantage also depends on the comparator; conservation of an integral does not establish superiority in a global velocity norm.

A surface-row-only norm confirms that the poor thin-cell result is not caused by pooling the bed row. At 512 cells and $q_s = 0.001$, its L error decreases from 0.0093744 to 0.00073369 under fourfold trace refinement, compared with an unchanged exact-field centroid error of 0.0001004. Conversely, for $q_s = 0.1$ on the finest original grid, the L surface-row error is 8.7735×10^{-5} , slightly below the control error. This local exception does not change the ranking of the pooled cut-cell norm in Figure 4.

A.5 Fixed overturning-wave state

Before the native-bed adjustment, the fixed wave state satisfies the offline volume and momentum checks on all three receiver grids (Table 10). Native loading on the matched geometry is reported separately below. The repeated momentum error reflects shared boundary data and does not establish convergence with receiver spacing. Near-boundary point probes and liquid-volume means measure different quantities.

Table 10: Offline transfer measurements on the pre-adjustment polygon geometry, before loading into the receiver. MAC divergence is the full Cartesian identity before native receiver loading. Probe errors are pointwise-reference relative L^2 measurements.

Grid	e_V	e_M	MAC divergence L^∞	Probe error
L10	$1.75 \cdot 10^{-16}$	$6.96 \cdot 10^{-8}$	$1.78 \cdot 10^{-15}$	$7.67 \cdot 10^{-3}$
L11	$1.75 \cdot 10^{-16}$	$6.96 \cdot 10^{-8}$	$3.55 \cdot 10^{-15}$	$1.29 \cdot 10^{-3}$
L12	$1.75 \cdot 10^{-16}$	$6.96 \cdot 10^{-8}$	$7.11 \cdot 10^{-15}$	$1.07 \cdot 10^{-3}$

Independent BIE volume integration was performed for 12 cells selected by geometry alone with $0.05 < q_K < 0.95$. The sampled relative velocity error is 7.2115×10^{-4} , while the reported quadrature disagreement is 1.5248×10^{-8} . Close evaluation uses source refinement and a common potential-gauge subtraction on both boundaries. This test excludes sliver cells and uses the geometry preceding the native-bed adjustment. We next assess the reconstructed native geometry.

A.6 Independent reference on native bed-cut cells

Changing the represented bed and applying the volume-matching surface shift also changes the reference-integration domain. It is invalid to use an interior BIE formula indiscriminately outside its source domain. We therefore check whether each native liquid polygon lies inside the declared source polygon, allowing an area discrepancy no greater than $10^{-12}h^2$. Of 8,495 native mixed cells, 3,479 satisfy this check and 5,016 do not; the largest excluded area is 1.74993×10^{-7} in one cell. Containment in the source polygon is a numerical screening condition, not a proof of containment in the continuously Fourier-refined BIE contour.

After checking that a centroid fan triangulates each simple polygon without leaving it, six cells are selected deterministically from each of the intervals $0.05 < q < 0.95$ and $0.95 < q < 1$. The eligible populations are 3,150 and 134 cells. All 12 selected cells intersect the bed, not the free surface; the source-domain screen excludes many displaced free-surface polygons. The sample therefore tests native bed-cut means and cannot establish accuracy over the whole free surface. The reference is the fixed source BIE field integrated over these native polygons; it is not a re-solved potential-flow problem on the receiver's modified bed.

Independent BIE velocity evaluation is integrated with a radial Duffy map on the centroid-fan triangles. The relative difference between three- and five-point Gaussian rules is 2.26482×10^{-10} at the original source-evaluation tolerance. Because agreement between volume-quadrature rules alone does not bound their common source-evaluation error, the five-point reference is recomputed on the same cells with 100-fold tighter source tolerances. This requires two successive source-refinement agreements within $10^{-9} + 10^{-7}\|u\|_\infty$ per point. All 1,225 quadrature points satisfy this criterion by a maximum source resampling factor of 8,192. The relative change of the integrated reference means is 6.81445×10^{-11} , well below the reconstruction errors. Both reconstructions are compared with this stricter five-point reference using the unweighted norm over the sampled cell means (Table 11).

Table 11: Independent reference for 12 selected native bed-cut cells. Errors are unweighted relative Euclidean norms of phase-mean velocities.

Liquid-fraction interval	Cells	L moments	Bilinear-centroid control
(0.05, 0.95)	6	4.82343×10^{-6}	1.77693×10^{-4}
(0.95, 1)	6	2.07174×10^{-5}	2.88663×10^{-5}
Pooled sample	12	1.61051×10^{-5}	1.17025×10^{-4}

The pooled L error is smaller by a factor of 7.27. It is smaller in nine individual cells, whereas the control is more accurate in three. This sampled comparison does not cover the thinnest liquid cells. An attempted four-cell BIE reference for thinner cells failed the close-evaluation criterion at 11 of 144 quadrature points, all in two cells with $q = 0.0003553$ and $q = 0.03266$; those points are excluded from the accuracy comparison. Thin cells are assessed with the analytic flat-boundary test.

A.7 Receiver geometry and startup

Table 12 isolates the effect of receiver-matched geometry. The original preparation had the prescribed offline liquid volume but lost liquid when loaded. The measured loss is explained by clipping against the different native solid geometry. Matching the receiver convention removes that discrepancy to roundoff for the tested state.

Table 12: Geometry consistency and liquid-volume loss at loading.

Measurement	Polygon-only	Receiver-matched
Maximum offline/native c_s difference	7.17409×10^{-4}	9.37028×10^{-14}
Relative initial mass error	1.19487×10^{-7}	1.75329×10^{-16}
Relative initial momentum error	8.89928×10^{-8}	6.96337×10^{-8}

The matched L12 input reaches $t = 0.01$ in 33 steps. At the final state the relative physical liquid-volume change is 8.76644×10^{-16} and the embedded volume-flux divergence L^∞ is 3.16902×10^{-12} . The reported liquid-regular and all-cell maximum speeds are 1.83611 and 2.52053. The former uses the mask $c_s \geq 0.5$, $f > 0.01$ and does not bound every small embedded cell. The measured wall time, 225.505 seconds, includes 21 checkpoint commits; it is not a transfer-only cost or a speedup comparison.

A.8 Paired native initialisations

Both L12 branches complete the exact common interval $0 \leq t \leq 0.01$. Their input phase moments differ: the L relative momentum error is 6.96337×10^{-8} , about 15.8 times smaller than the uncorrected bilinear-centroid control’s error. This improves the initial integral constraint; subsequent accuracy requires an independent evolved reference. Table 13 reports the branch differences at the initial and final common times.

Table 13: Difference between the two native branches at identical times. These are sensitivity measurements, not errors against an exact flow.

Measurement	$t = 0$	$t = 0.01$
Liquid velocity difference D_u	9.37400×10^{-4}	9.48721×10^{-4}
VOF fraction difference D_f	0	4.37319×10^{-5}
Column liquid-depth difference D_H	0	5.80190×10^{-6}
Relative liquid-volume difference	0	5.25987×10^{-16}

At $t = 0.01$, the density-normalised liquid kinetic energies $\frac{1}{2}h^2 \sum_K q_K \|u_K\|_2^2$ are 0.43967054 and 0.43969504 for the L and control branches. The maximum column-depth difference is 1.16104×10^{-4} . The receiver thus retains some sensitivity to the input moment operator, while liquid volume agrees to roundoff. An independent evolved reference is needed to rank their accuracy.

A.9 Time-step sensitivity

A second paired experiment halves the initial step limit from 0.00125 to 0.000625, the maximum step from 0.000625 to 0.0003125, and both CFL controls from 0.05 to 0.025. Each full-step branch takes 32 steps and each half-step branch takes 64 steps to reach $t = 0.01$. These runs use a different step schedule from the 33-step startup above. Table 14 uses the same relative norms, with the first-listed state supplying the denominator and liquid weights.

Table 14: Endpoint sensitivity to one time-step halving.

Comparison at $t = 0.01$	D_u	D_f
Full-step L versus control	9.48721×10^{-4}	4.37319×10^{-5}
Half-step L versus control	9.58218×10^{-4}	4.28176×10^{-5}
L full versus half step	5.13667×10^{-3}	1.21956×10^{-4}
Control full versus half step	5.15092×10^{-3}	1.22843×10^{-4}

The branch differences change little under this halving, but the full-versus-half-step differences are larger than the differences between initialisations. Thus the experiment detects a persistent short-time influence of the transfer operator without establishing temporally resolved dynamic superiority. Two time settings do not establish a convergence order or an error against the physical trajectory.

A.10 Propagation and boundary-trace resolution

The smooth native benchmark separates a global initial norm from the later influence of that error. With the original trace factor, L transfer has a smaller initial global error on the 512×32 grid, but its error at $0.1T$ is 2.88 times the bilinear control’s error. On 1024×64 , the initial L error is already 4.31 times larger and the endpoint error is 10.19 times larger. Thus a small initial global norm, like a closed global moment balance, does not ensure a better subsequent state. Figure 5 compares the corresponding error histories and the effect of increasing the trace density. Its vertical axes are logarithmic.

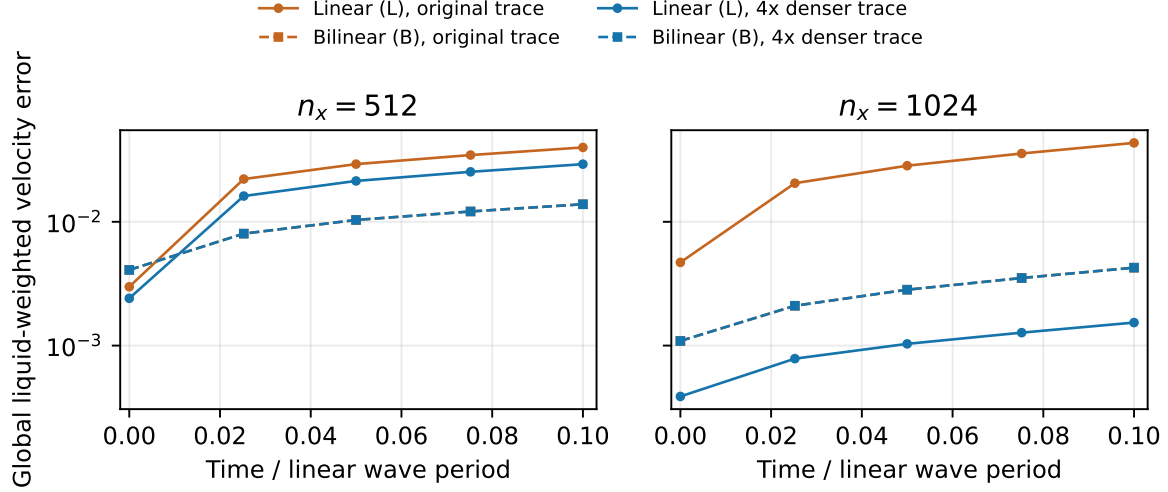


Figure 5: Native propagation of initial transfer error for the smooth-wave benchmark, using the full time-step limit. Each curve is measured against the same-grid native branch initialised with exact liquid-cell means. The two trace densities give practically coincident bilinear-control curves. The refined L transfer improves the finer-grid result but still loses to the control on the coarse grid. This is not an error plot against an exact two-phase trajectory.

The trace-density intervention keeps source values, geometry, gas fields and reference inputs fixed. The computed streamfunction changes by only 4.65×10^{-13} and 1.39×10^{-12} in relative norm at the two grids, while a L wet-cell velocity component changes by as much as 0.0136 ; recomputed preprocessing MAC face velocities change by at most 2.15×10^{-12} . Together with the unchanged liquid fractions, this localises the main change to boundary-trace and edge-moment reconstruction, including trace values at boundary intersections, rather than geometry or the bulk harmonic solve. The data do not isolate physical-edge integrals from all other boundary-dependent edge terms. In the original 1024×64 input, 99.43% of the L squared liquid-weighted velocity error is in the upper thin liquid row.

On 1024×64 , fourfold trace refinement lowers the L initial error from 4.69958×10^{-3} to 3.8820×10^{-4} . At $0.1T$, its global velocity error is 1.53448×10^{-3} , compared with 4.25970×10^{-3} for the same-streamfunction bilinear control: a 2.78-fold reduction. The relative VOF errors are 5.31475×10^{-6} and 2.14689×10^{-5} . Normalising the column-depth error by the reference wave component $\|H^R - \langle H^R \rangle\|_2$, rather than the background depth, gives 1.34113×10^{-4} and 5.41771×10^{-4} , a 4.04-fold reduction. The maximum absolute column-depth differences are 2.518×10^{-6} and 9.111×10^{-6} . The reference wave’s mode-16 amplitude is 0.01396 . These measurements demonstrate a reduced propagated transfer error in this finer-grid case.

The coarse-grid result remains unfavourable: refined L and control velocity errors are 2.92106×10^{-2} and 1.38609×10^{-2} . The density intervention therefore does not establish a grid-independent advantage or a uniform small-cell accuracy guarantee.

The favourable 1024×64 comparison survives one time-step halving: the L and control velocity errors become 1.59996×10^{-3} and 4.46877×10^{-3} , giving a 2.79-fold reduction. Their wave-normalised column-depth errors are 1.48757×10^{-4} and 6.34783×10^{-4} , a 4.27-fold reduction. Recorded step counts are 210 and 420, with observed maximal steps 0.0025 and 0.00125. The corresponding unrefined-trace comparison remains unfavourable by a factor of 10.18.

The transfer ranking persists under this step halving, but temporal convergence of the exact-input reference is not established. Its full-versus-half-step velocity difference is 2.04241×10^{-2} , larger than the L-versus-reference differences at either time setting. The primary ratios therefore quantify the propagation of initialisation error under a common numerical evolution, not the total error against a physical wave solution.

A further comparison uses the half-step exact-input branch as the common reference for both full-step candidates. Their velocity discrepancies are 0.0209794 (L) and 0.0193803 (control): the L result is 8.3% worse. Thus the favourable matched-step ratios cannot be promoted to an assertion of improved total trajectory accuracy. Common temporal error and its interaction with initialisation dominate this cross-step comparison. On 512×32 with the refined trace, the cross-step velocity ranking also changes: discrepancies are 0.0204883 (L) and 0.0223687 (control), whereas wave-normalised column-depth discrepancies are 0.00362516 and 0.00354270. The matched-step transfer ranking therefore does not extend to this cross-step comparison.

The ranking reversal can be resolved algebraically without treating the half-step trajectory as an exact physical reference. Let $u_m^{\Delta t}$ denote a transferred branch, $u_*^{\Delta t}$ the exact-input branch, and $u_*^{\Delta t/2}$ the latter branch at the smaller step. With a single liquid-volume inner product weighted by the half-step reference, define $e_m = u_m^{\Delta t} - u_*^{\Delta t}$ and

Table 15: Global propagated transfer errors at $t = 0.1T$. The trace factor is the number of trace samples per original source segment. Ratios $R_u = E_u^B/E_u^L$ and R_H for wave-normalised column depth exceed one when Linear (L) is better than Bilinear (B). Every comparison uses a reference with the same grid and time-step limit.

n_x	Trace	$10^3 \Delta t_{\max}$	E_u^L	E_u^B	R_u	R_H
512	8	2.5	0.03988	0.01386	0.35	0.64
512	8	1.25	0.04084	0.01419	0.35	0.60
1024	8	2.5	0.04340	0.00426	0.10	0.13
1024	8	1.25	0.04549	0.00447	0.10	0.14
512	32	2.5	0.02921	0.01386	0.47	0.84
1024	32	2.5	0.00153	0.00426	2.78	4.04
1024	32	1.25	0.00160	0.00447	2.79	4.27
2048	64	2.5	0.00051	0.00175	3.44	3.51

$b = u_*^{\Delta t} - u_*^{\Delta t/2}$. Then

$$\|u_m^{\Delta t} - u_*^{\Delta t/2}\|^2 = \|e_m\|^2 + \|b\|^2 + 2\langle e_m, b \rangle. \quad (26)$$

At the common endpoint of the refined 1024×64 case, the normalised transfer norms are 0.00153846 and 0.00427009 for L and control inputs, while $\|b\|/\|u_*^{\Delta t/2}\| = 0.02042411$. The corresponding error-vector cosines are $+0.3282$ and -0.3427 . Thus the smaller control discrepancy against the half-step reference includes cancellation of temporal error; it does not by itself demonstrate a more accurate physical trajectory. The squared-norm identity holds for the endpoint fields to an absolute normalised residual below 4.4×10^{-19} . These norms differ slightly from the matched-step values because their weights and normalising reference are held fixed to the half-step branch.

The 2048×128 joint-refinement check gives the same matched-step ranking. Initial L and control errors are 1.03094×10^{-4} and 3.34981×10^{-4} . At $0.1T$, their global velocity errors are 5.08178×10^{-4} and 1.74890×10^{-3} , a 3.44-fold reduction. Wave-normalised column-depth errors are 4.7457×10^{-5} and 1.66515×10^{-4} , a 3.51-fold reduction. All three branches complete 210 steps; maximum recorded post-start embedded-volume flux divergence is 8.10×10^{-14} and recorded mass drift is zero at output precision. This supports the matched-setting transfer benefit at a second spatial resolution, but supplies no additional time-step convergence evidence. Table 15 retains the complete tested matrix.

A.11 Earlier Matched continuations and time-step control

These earlier continuations use the original common receiver build. They compare M with B and exact-input branches, retaining the source resolution and trace factor 8. At 512×32 and 1024×64 , geometry, volume fractions, supplied face data and gas velocities are identical to those of the exact-input branches. The solver executable, library, physical coefficients and full-step settings are unchanged. Both runs complete 210 steps to the same endpoint $t = 0.1T$. The receiver reconstructs its initial face fluxes from each branch's cell velocities; the common supplied face arrays are not loaded as prescribed runtime fluxes.

At 512×32 , the global velocity discrepancy against the exact-input trajectory is 0.00175849, compared with 0.01386093 for B: a 7.88-fold reduction. The wave-normalised surface-profile error decreases from 2.55281×10^{-3} to 3.92627×10^{-5} , a 65.0-fold reduction. At 1024×64 , velocity errors are 0.000434603 and 0.004259703, a 9.80-fold reduction, while surface-profile errors are 7.33856×10^{-6} and 5.41771×10^{-4} , a 73.8-fold reduction. Thus M improves both subsequent observables at the original trace density. The favourable comparison holds at every recorded nonzero time, not only at the endpoint. Doubling the grid resolution reduces the matched global velocity error by factors 4.015 initially and 4.046 at the endpoint, corresponding to two-grid observed orders 2.005 and 2.017. These ratios are consistent with second-order behaviour over the tested refinement pair; they are not a multi-level asymptotic convergence study of the receiver.

Halving the time step on 1024×64 preserves the improvement. The matched run completes 420 steps and gives velocity error 0.0004361803 against the half-step exact-input branch, versus 0.0044687721 for the half-step B control (10.25-fold reduction). The respective surface-profile errors are 7.65566×10^{-6} and 6.34783×10^{-4} (82.9-fold reduction). The common receiver settings are held fixed within each comparison; these ratios measure propagated transfer error rather than the receiver's temporal truncation error.

The norms use $q_{\text{ref}} = f_{\text{ref}} c_{s,\text{ref}}$ and $H = h \sum_j f_{ij} c_{s,ij}$ from the output fields. Equal endpoint times and common inputs isolate the transfer operator within each receiver discretisation.

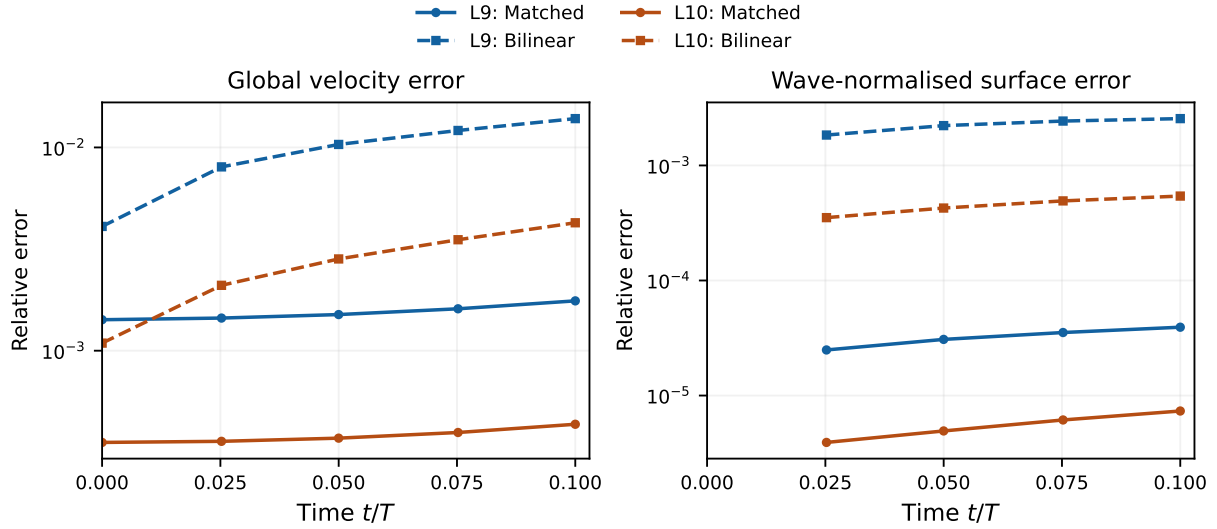


Figure 6: Propagation of transfer error with Matched (M) and Bilinear (B) initialisations. L9 and L10 denote 512×32 and 1024×64 grids, both with $\Delta t_{\max} = 0.0025$. Each curve uses the exact-input numerical trajectory at its own grid and identical time-step settings as the reference. The surface error is undefined at the initially flat state and is shown only at nonzero times.

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